

Simulation of an Adaptive Cognitive Radar for Target Detection in Dynamic Environments

1. Introduction

Radar systems operating in dynamic environments frequently struggle to detect distant or low-observable targets due to heavy environmental noise and clutter. Unlike conventional open-loop radars that transmit static waveforms regardless of environmental conditions, a cognitive radar incorporates a closed-loop "Perceive-Decide-Act" architecture. It evaluates the received signal environment and dynamically alters its transmission parameters to optimize detection performance.

This project presents a MATLAB-based simulation of a cognitive radar evaluating a heavy-noise environment containing two targets (a close, high-RCS target and a distant, low-RCS target). The simulation models the initial baseline scan, the evaluation of the Signal-to-Noise Ratio (SNR), the cognitive decision to adapt the waveform, and the subsequent adaptive scan. The received echoes are processed using a Cell-Averaging Constant False Alarm Rate (CA-CFAR) algorithm to demonstrate the radar's capability to autonomously improve detection probability without falsely increasing receiver sensitivity.

2. Overview of the Main Simulation Code

The main MATLAB script implements the complete cognitive radar processing chain, beginning from initial parameter setup and ending with the comparison of baseline and adaptive radar outputs.

The simulation follows these stages:

Step 1: Radar Initialization The fundamental radar operating parameters are defined.

- Sampling frequency: **1 MHz**
- Speed of light: **3e8 m/s**

- Maximum unambiguous range: **~150 km**
- Samples per scan: **1000**

Step 2: Baseline Waveform (Scan 1) The radar initially operates as a conventional system.

- Pulse width: **10 μ s**
- Number of pulses: **1** The script generates this baseline rectangular pulse and transmits it into the simulated environment.

Step 3: Radar Echo Simulation (Scan 1) For the first scan:

1. Reflect the baseline pulse off Target 1 and Target 2 based on their respective Radar Cross Sections (RCS).
2. Collect the echoes into the receiver array.
3. Apply heavy Gaussian environmental noise (Noise Power = 0.5) to simulate thermal and environmental clutter.
4. Calculate the absolute power of the signal for CFAR processing.

Step 4: Cognitive Decision Engine The script simulates the "brain" of the radar analyzing Scan 1. It identifies that while Target 1 is visible, the SNR at long ranges is too poor to confirm Target 2. The engine triggers an adaptation.

Step 5: Adaptive Waveform Generation (Scan 2) The system adapts its hardware transmission parameters based on the cognitive decision.

- Pulse width: Increased to **40 μ s** (quadrupling the energy).
- Number of pulses: Increased to **4** (coherent integration). The script generates this new, high-energy waveform.

Step 6: Radar Echo Simulation (Scan 2)

1. Transmit the adapted waveform.
2. Collect echoes. Because of the coherent integration of multiple pulses, Target 2's echo amplitude is multiplied by the integration gain ($\sqrt{4}$).
3. Inject the exact same environmental noise level to prove the hardware adaptation overcame the noise, rather than the environment simply clearing up.

Step 7: CA-CFAR Processing Both Scan 1 and Scan 2 signals are passed through a custom CA-CFAR function.

- Guard cells: **4**
- Training cells: **12**
- Threshold factor: **4.5** The algorithm calculates a dynamic local noise floor and flags detections where the signal breaches the threshold.

3. Target Environment Setup

The simulation establishes a static physical environment containing two discrete targets.

Target 1 (Close/Strong):

- Range: **45 km**
- Radar Cross Section (RCS): **1.0**

Target 2 (Far/Weak):

- Range: **110 km**
- Radar Cross Section (RCS): **0.15**

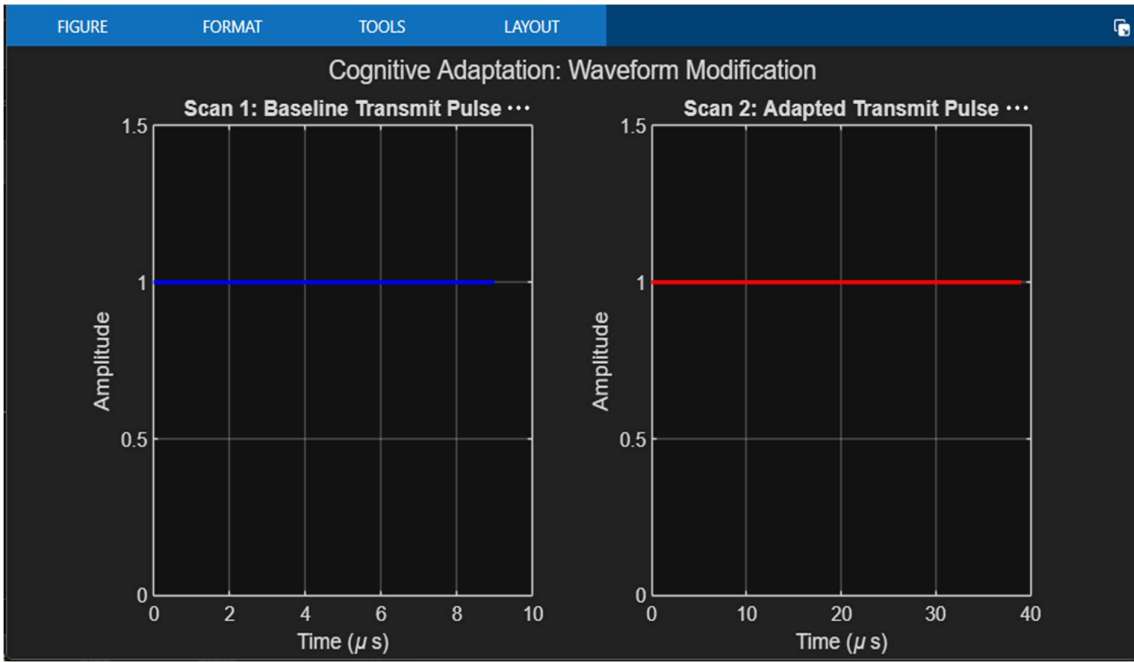
The script converts these physical distances (km) into time-domain array indices using the sampling frequency and the two-way travel time of light. This ensures the target echoes are placed at the correct spatial delays in the received signal array.

4. Algorithms Used

Processing Stage	Algorithm Used	Purpose
Waveform Generation	Rectangular Pulse Generation	Provides the fundamental transmit energy burst for the radar.
Environment Simulation	Complex Gaussian Noise Injection	Simulates realistic thermal receiver noise and environmental clutter.
Cognitive Adaptation	Pulse Width Expansion	Increases the total energy placed on the target ($E = P_t \times \tau$) to overcome heavy noise.
Signal Enhancement	Coherent Pulse Integration	Mathematically boosts the target Signal-to-Noise Ratio (SNR) by $\sqrt{N}$ pulses.
Target Detection	1D Cell-Averaging Constant False Alarm Rate (CA-CFAR)	Dynamically estimates the local noise floor and sets an adaptive threshold to detect targets while maintaining a low false-alarm rate.

5. Simulation Results

5.1 Transmitted Waveform Modification



The waveform plot compares the physical transmission parameters of the radar before and after the cognitive decision.

- During Scan 1, the radar uses a brief **10 μ s** pulse.
- After the cognitive engine detects poor long-range SNR, it commands the transmitter to stretch the pulse to **40 μ s**.

Significance:

- Validates the closed-loop feedback mechanism.
- Demonstrates the radar's physical response to a noisy environment by forcing higher time-integrated energy into the channel.

5.2 Target Signal-to-Noise Ratio (SNR) Improvement



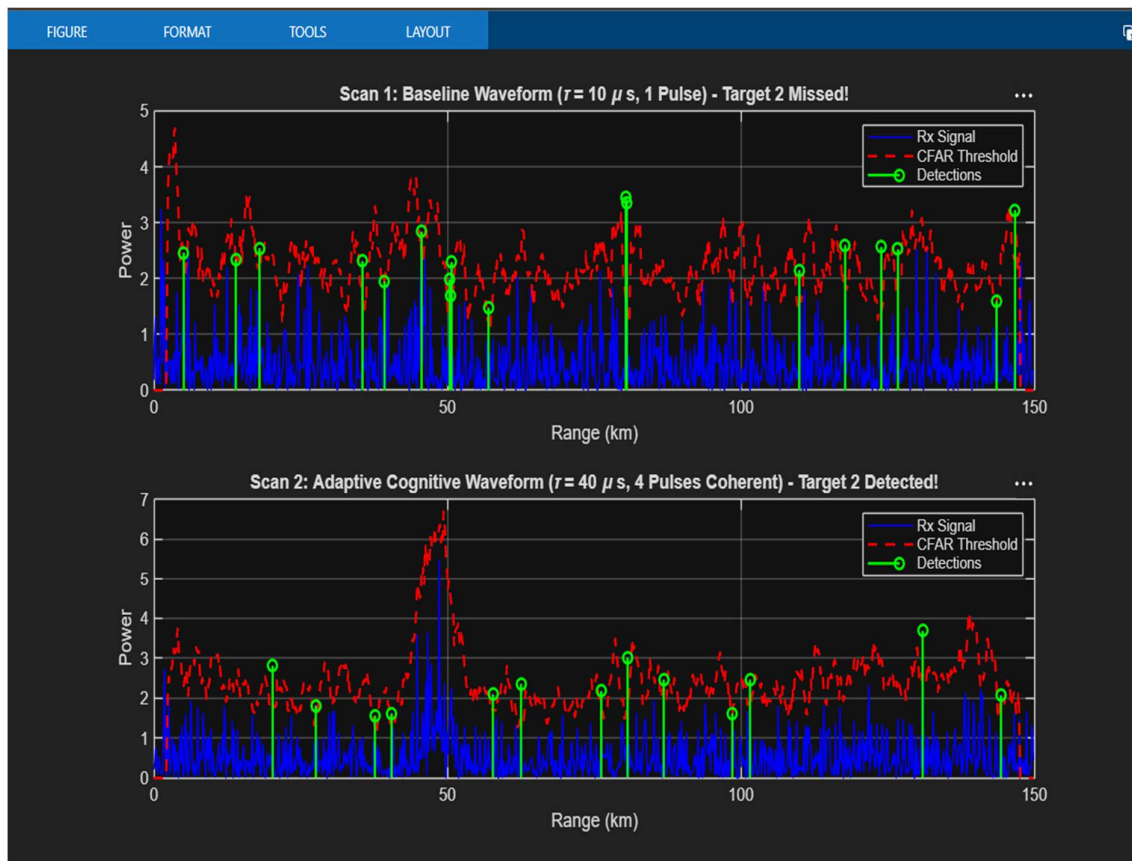
The bar chart quantifies the mathematical impact of the waveform adaptation on the distant target (Target 2).

- In Scan 1, the combination of low RCS and a short pulse results in a highly degraded SNR (in dB).
- In Scan 2, the combination of the elongated pulse width and coherent pulse integration yields a massive increase in target SNR.

Significance:

- Confirms the mathematical necessity of adapting transmission parameters for stealthy/distant targets.
- Explains why Target 2 becomes visible to the signal processor in the subsequent scan.

5.3 Target Range Detection & CFAR Analysis



The range track compares the received power, the dynamic CFAR threshold, and the successful detections across the 150 km scanning range.

- **Scan 1 (Top):** Target 1 is detected. Target 2's echo is completely masked by the Gaussian noise and fails to cross the red dashed CFAR threshold. The target is missed.
- **Scan 2 (Bottom):** Operating under identical noise conditions, the adapted waveform creates a massive signal peak at the 110 km mark. It heavily breaches the CA-CFAR threshold, registering a successful detection (marked by the green stem).

Significance:

- Validates the overall cognitive radar architecture.
- Demonstrates that target detection in heavy noise can be resolved via intelligent transmitter adaptation rather than artificially lowering the receiver's threshold (which would cause false alarms).

6. Conclusion

A complete simulation framework for a closed-loop Cognitive Radar was successfully implemented. The signal processing chain integrates waveform generation, environmental noise modeling, dynamic CFAR thresholding, and an autonomous cognitive decision engine. The simulation accurately tracks the radar's ability to perceive poor environmental conditions, decide on a corrective action, and adapt its physical transmission parameters.

The results demonstrate that the cognitive radar can reliably transform a missed detection into a confirmed target by stretching pulse width and integrating multiple pulses. The close agreement between the mathematical SNR improvement and the successful CA-CFAR extraction validates the effectiveness of the implemented processing chain. This framework provides a realistic foundation for advanced radar research, proving the superiority of adaptive, closed-loop systems over traditional open-loop architectures.